⛈️ The Strike & Staying Safe: Guided Notes

Chemistry · Unit 1C · Level 3 (closer to the class work) · goes with L1.10-L1.14 (Energy Flow, The Strike, Conductors & Insulators, No Swimming, Unit Test)

Name: _____

Use the class slides and your reading. Key words:

potential energy | *thermal energy* | *stepped leader* | *return stroke* | *conductor* | *insulator* | *ion* | *current* | *dissolve* | *lightning rod*

A. Energy flow (L1.10)

1. Separated charge stores _____ energy.

2. List the three forms of energy that potential energy changes into during a strike.

_____

B. The strike (L1.11)

3. Fill in the table.

Step	What happens
1. Stepped leader	
2. Streamer	
3. Return stroke	
4. Thunder	

C. Conductors and insulators (L1.12, L1.13.5)

4. In a _____, electrons move easily. In an _____, they do not.

5. Why are metals good conductors? Use the word electrons.

_____

_____

6. Fill in the table.

Material	Conductor or insulator?
copper wire	
rubber	
salt water	
dry wood	

D. Salt water and safety (L1.13, L1.14)

7. **Reading the No Swimming data: 1 amp (A) = 6.28×10^{18} charges per second.**

> Example: 2 A = $2 \times 6.28 \times 10^{18} = 12.56 \times 10^{18} = 1.256 \times 10^{19}$ charges per second.

a) 0.5 A = ____ $\times 10^{18}$ charges per second _____

b) 3 A = ____ $\times 10^{18}$ charges per second _____

8. Claim: How does dissolving salt in water affect the current? Give evidence from the lab and reasoning.

9. Draw a house with a lightning rod. Use arrows to show the path of the charge.

⛈️ The Strike & Staying Safe: Reading

Chemistry · Unit 1C · Level 3 (closer to the class work) · goes with L1.10-L1.14 (Energy Flow, The Strike, Conductors & Insulators, No Swimming, Unit Test)

Name: _____

The Strike and Staying Safe

When charge separates in a storm cloud, energy is stored, a bit like a stretched rubber band. This is called potential energy. The more charge that separates, the more energy is stored. When the charge finally moves, the stored energy changes into heat (thermal energy), light and sound.

A strike happens in steps. First, a faint, zig-zag channel of negative charge called a stepped leader moves down from the cloud. As it gets close, positive streamers reach up from tall objects like trees, towers and people. When a leader and a streamer meet, a huge current flows. That bright flash is the return stroke. It heats the air so fast that the air expands with a boom: thunder.

Some materials let charge move through them easily. These are conductors. Metals are good conductors because some of their electrons are free to move from atom to atom. Other materials, like rubber, plastic, glass and dry wood, hold their electrons tightly. These are insulators.

Pure water does not conduct well, but salt water does. When salt dissolves, it breaks into ions: charged particles that can move. In the No Swimming lab, the light bulb got brighter as more salt was added, because more ions carried more current. Pools, lakes and the ocean all have dissolved ions, so lightning can travel through them.

That is why some places are safer than others. A lightning rod is a metal conductor that gives the charge a safe path into the ground. Buildings with wiring and plumbing, and hard-top cars, are safe places. Open fields, water and tall, lone trees are dangerous. Remember: when thunder roars, go indoors.

Check your reading

1. What energy changes happen during a lightning strike?

2. Why is salt water a conductor but pure water is not?

3. Explain how a lightning rod protects a house.

💡 Conductivity Lab: What Lets Electricity Through?: Lab Sheet

Chemistry · Unit 1C · Level 3 · goes with L1.13 No Swimming, L1.13.5 You Light Up My Life

Name: _____

Question: How does dissolving a substance in water affect whether it conducts electricity?

Materials

conductivity tester (low voltage), foil or copper, wax, sugar, salt, distilled water, cups, balance and graduated cylinder

⚠️ Safety

- Goggles on the whole lab.
- Never taste anything, even salt or sugar.
- Only the low-voltage tester. Nothing in wall outlets.
- Wash hands after.

Prediction

Predict which materials will conduct. Explain using electrons or ions.

_____

_____

Procedure

1. Check that the tester bulb lights when its two ends touch.
2. Test each solid: foil/copper, wax, sugar, salt crystals. Record.
3. Make 100 mL salt water at 1%, 5% and 10% (1 g, 5 g, 10 g of salt). Test each.
4. Test distilled water and sugar water.
5. Dry the tester between every test.
6. Sort the materials into conductors and insulators.

Data

Material	Prediction	Bulb: bright / dim / off	Conductor or insulator?
foil / copper			
wax			
salt crystals			
distilled water			
1% salt water			
10% salt water			
sugar water			

Analysis (Claim, Evidence, Reasoning)

1. Claim: How does dissolving salt in water affect conductivity?

2. Evidence: Use your data for distilled water, 1% and 10% salt water.

3. Reasoning: Why does sugar water NOT conduct? Use the word ions.

4. **Current to charges: 1 A = 6.28×10^{18} charges per second.**

> Example: 0.2 A = $0.2 \times 6.28 \times 10^{18} = 1.256 \times 10^{18}$ charges per second.

a) 0.5 A = ____ $\times 10^{18}$ charges/s _____

b) Which moves more charge each second: 0.1 A or 0.4 A? _____

⛈️ The Strike & Staying Safe: Quick Check

Chemistry · Unit 1C · Level 3 (closer to the class work) · goes with L1.10-L1.14 (Energy Flow, The Strike, Conductors & Insulators, No Swimming, Unit Test)

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. During a strike, stored potential energy changes mostly into…

A. Chemical energy | B. More charge | C. Heat, light and sound

2. The bright flash of lightning is the…

A. Return stroke | B. Stepped leader | C. Insulator

3. Which material is an insulator?

A. Copper | B. Salt water | C. Rubber

4. In the No Swimming lab, adding more salt made the current…

A. Increase | B. Decrease | C. Stay at zero

5. Explain why salt water conducts electricity. Use the word ions.

6. Why is a hard-top car or a building safer than an open field in a lightning storm?

⛈️ The Strike & Staying Safe: Guided Notes · ANSWER KEY

Chemistry · Unit 1C · Level 3 (closer to the class work) · goes with L1.10-L1.14 (Energy Flow, The Strike, Conductors & Insulators, No Swimming, Unit Test)

Name: _____

Use the class slides and your reading. Key words:

potential energy · *thermal energy* · *stepped leader* · *return stroke* · *conductor* · *insulator* · *ion* · *current* · *dissolve* · *lightning rod*

A. Energy flow (L1.10)

1. Separated charge stores **potential** energy.

2. List the three forms of energy that potential energy changes into during a strike.
Thermal (heat), light, and sound.

B. The strike (L1.11)

3. Fill in the table.

Step	What happens
1. Stepped leader	**Negative charge zig-zags down from the cloud**
2. Streamer	**Positive charge reaches up from a tall object**
3. Return stroke	**They connect; huge current, bright flash**
4. Thunder	**Air heats and expands fast: boom**

C. Conductors and insulators (L1.12, L1.13.5)

4. In a **conductor**, electrons move easily. In an **insulator**, they do not.

5. Why are metals good conductors? Use the word electrons.
Some of their electrons are free to move from atom to atom.

6. Fill in the table.

Material	Conductor or insulator?
copper wire	**conductor**
rubber	**insulator**
salt water	**conductor**
dry wood	**insulator**

D. Salt water and safety (L1.13, L1.14)

7. **Reading the No Swimming data: 1 amp (A) = 6.28×10^{18} charges per second.**

> Example: 2 A = $2 \times 6.28 \times 10^{18} = 12.56 \times 10^{18} = 1.256 \times 10^{19}$ charges per second.

a) 0.5 A = ____ $\times 10^{18}$ charges per second **3.14**

b) 3 A = ____ $\times 10^{18}$ charges per second **18.84 (= 1.884×10^{19})**

8. Claim: How does dissolving salt in water affect the current? Give evidence from the lab and reasoning.
More salt → more current (brighter bulb, higher amps) because dissolved salt makes more ions that carry charge.

9. Draw a house with a lightning rod. Use arrows to show the path of the charge.
Rod on the roof, wire down the side, arrows from cloud → rod → wire → ground.

🔑 *Key written by our team from Mr. Brittle's worksheets and slides, please check.*

💡 Conductivity Lab: What Lets Electricity Through?: Lab Sheet · ANSWER KEY

Chemistry · Unit 1C · Level 3 · goes with L1.13 No Swimming, L1.13.5 You Light Up My Life

Name: _____

Question: How does dissolving a substance in water affect whether it conducts electricity?

Materials

conductivity tester (low voltage), foil or copper, wax, sugar, salt, distilled water, cups, balance and graduated cylinder

⚠️ Safety

- Goggles on the whole lab.
- Never taste anything, even salt or sugar.
- Only the low-voltage tester. Nothing in wall outlets.
- Wash hands after.

Prediction

Predict which materials will conduct. Explain using electrons or ions.

Metal (free electrons) and salt water (ions) will conduct; wax, sugar, solid salt and pure water will not (or only very little).

Procedure

1. Check that the tester bulb lights when its two ends touch.
2. Test each solid: foil/copper, wax, sugar, salt crystals. Record.
3. Make 100 mL salt water at 1%, 5% and 10% (1 g, 5 g, 10 g of salt). Test each.
4. Test distilled water and sugar water.
5. Dry the tester between every test.
6. Sort the materials into conductors and insulators.

Data

Material	Prediction	Bulb: bright / dim / off	Conductor or insulator?
foil / copper		**bright**	**conductor**
wax		**off**	**insulator**
salt crystals		**off**	**insulator**
distilled water		**off**	**insulator**
1% salt water		**dim**	**conductor**
10% salt water		**bright**	**conductor**
sugar water		**off**	**insulator**

Expected results (your class data may differ).

Analysis (Claim, Evidence, Reasoning)

1. Claim: How does dissolving salt in water affect conductivity?
Dissolving salt makes water conduct; more salt conducts more.

2. Evidence: Use your data for distilled water, 1% and 10% salt water.
Distilled water: off; 1%: dim; 10%: bright.

3. Reasoning: Why does sugar water NOT conduct? Use the word ions.
Sugar dissolves as neutral molecules, not ions, so nothing charged can move to carry current.

4. **Current to charges: 1 A = 6.28×10^{18} charges per second.**

> Example: 0.2 A = $0.2 \times 6.28 \times 10^{18} = 1.256 \times 10^{18}$ charges per second.

a) 0.5 A = ____ $\times 10^{18}$ charges/s **3.14**

b) Which moves more charge each second: 0.1 A or 0.4 A? **0.4 A (4 times as much)**

🔑 *Key written by our team from Mr. Brittle's lab sheets, please check.*

⛈️ The Strike & Staying Safe: Quick Check · ANSWER KEY

Chemistry · Unit 1C · Level 3 (closer to the class work) · goes with L1.10-L1.14 (Energy Flow, The Strike, Conductors & Insulators, No Swimming, Unit Test)

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. During a strike, stored potential energy changes mostly into…

A. Chemical energy | B. More charge | **C. Heat, light and sound**

2. The bright flash of lightning is the…

A. Return stroke | B. Stepped leader | C. Insulator

3. Which material is an insulator?

A. Copper | B. Salt water | **C. Rubber**

4. In the No Swimming lab, adding more salt made the current…

A. Increase | B. Decrease | C. Stay at zero

5. Explain why salt water conducts electricity. Use the word ions.

Salt dissolves into ions, charged particles that move and carry current.

6. Why is a hard-top car or a building safer than an open field in a lightning storm?

The metal/wiring conducts the charge around you and into the ground; in a field you may be the tallest object.

🔑 *Reworded from his L1.9 review and L1.14 practice test, please check.*